

Is CMV PCR of inner ear fluid during cochlear implantation a way to diagnose CMV-related hearing loss?

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To evaluate the diagnostic performance of cytomegalovirus (CMV) polymerase chain reaction (PCR) on inner ear fluid collected during cochlear implantation and to assess its interest in current practice. This monocentric prospective study included consecutive children presenting with severe to profound sensorineural hearing loss (SNHL) who were candidates for unilateral and/or bilateral cochlear implantation. The etiology of the SNHL was determined before cochlear implantation when possible. During the surgery, drop-like samples of inner ear fluid and blood were collected. CMV PCR was then performed on both samples. Between January 2017 and September 2021, 113 children with severe to profound SNHL underwent cochlear implantation with inner ear fluid collection. Among these children, 77 of them presented with a known cause of SNHL (68%) and 36 of them had an unknown cause of SNHL at the time of surgery (32%). Sensitivity and specificity of the CMV PCR on inner ear fluid were 60% (95% CI: [49-71]) and 98% (95% CI: [96-100]), respectively. Positive and negative predictive values were 90% (95% CI: [83-97]) and 92% (95% CI: [86-98]), respectively. A sensitivity analysis according to age at cochlear implantation showed a decrease with age. Sampling of inner ear fluid during cochlear implant surgery is an interesting, simple and safe way to diagnose CMV-related hearing loss, especially when the diagnosis of congenital infection can no longer be confirmed. However, the sensitivity decreases with age. NCT04724265 What is Known: • Congenital cytomegalovirus infection is the leading infectious cause of neurological disabilities and sensorineural hearing loss in children. In the absence of systematic screening at birth, many cCMV infections go undetected and are often undiagnosed despite the development of sensorineural sequelae. • Nearly 40% of indications for cochlear implantation are of unknown etiology. • Performing CMV PCR on inner ear fluid at the time of cochlear implantation is a safe way with high diagnostic performance (PPV = 90%, NPV = 92%) to detect a CMV-related hearing loss. • This sample may be interesting in cases of unknown cause of hearing loss in order to identify undiagnosed cCMV infections.

Cytomegalovirus infection in a single-centre Australian neonatal cohort.

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Congenital cytomegalovirus (cCMV) is the most common infectious cause of congenital malformation, non-genetic sensorineural hearing loss and neurodevelopmental sequelae in childhood. The primary aim of this retrospective cohort study was to identify the birth and neurodevelopmental outcomes of neonates diagnosed with symptomatic and asymptomatic cCMV in a large regional tertiary referral hospital. This was a retrospective cohort study of laboratory-based cCMV diagnoses in neonates born at a single study centre between January 2005 and January 2020. Audit of medical records was undertaken to evaluate maternal characteristics, symptom patterns, radiological and neurodevelopmental outcomes of neonates meeting the laboratory diagnostic criteria during the first 24 months. There were 45 neonates with proven CMV infection and 27 mothers with proven infection with an associated pregnancy outcome. Nineteen neonates were born at term (>37 weeks). Of these, 32 (71.1%) neonates had a significant intercurrent comorbidity and 22 (48.9%) neonates were reported to have a degree of delay in one or more developmental domains. A large proportion (77.3%) of the symptomatic untreated neonates had an unknown history of maternal infection compared to the asymptomatic (10.0%) and symptomatic treated (53.8%) neonates ($P = 0.001$). Up to half of the neonates with cCMV were at risk of developing a degree of developmental delay at our centre. Whether these outcomes are related primarily to CMV infection or

are confounded by the co-existence of prematurity is unclear and needs further evaluation in prospective studies.

Treatment of Congenital Cytomegalovirus and Ganciclovir Therapeutic Drug Monitoring in Twin Preterm Infants.

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Congenitally acquired cytomegalovirus (CMV) infection is the most prevalent congenital infection worldwide and the most frequent cause of acquired sensorineural hearing loss. The burden of the disease is even more important in premature and very low birth weight infants. However, few data exist on the treatment with intravenous ganciclovir and oral valganciclovir in this vulnerable population. We report the case of twins congenitally infected with CMV and born prematurely at 27 weeks' gestation. Treatment regimens were initially individualized for their prematurity and renal function, and then adjusted with therapeutic drug monitoring (TDM) to adapt to their continuously evolving physiologic maturation. As infants were aging, the plasmatic half-life of ganciclovir slowly decreased to term infant values around 10 weeks of chronological age, or 37 weeks of postmenstrual age. Results for blood polymerase chain reaction tests became negative and long-term follow-ups were satisfactory in both twins. The limited data for infants born before 32 weeks of gestation or at less than 1200 g and evolution of ganciclovir pharmacokinetic parameters justify the use of TDM in these settings.

A Review of Sensorineural Hearing Loss in Congenital Cytomegalovirus Infection.

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The intrauterine transmission of the TORCH group (toxoplasmosis, rubella cytomegalovirus (CMV), herpes simplex, and HIV) produces severe complications in the fetus, leading to major life challenges for the newborns. CMV is considered one of the leading causes of congenital hearing loss in babies born to infected mothers. The majority of the cases are asymptomatic but a certain proportion show symptoms, which may not be present until later. The human cytomegalovirus (HCMV) infection can also be transferred to the baby postnatally through breastfeeding. Mothers facing primary infection from the CMV have a greater tendency to transmit the infection to the fetus, whereas secondary infection due to reactivation of the virus in females who were affected before pregnancy, delivers asymptomatic babies in most cases. Vertical transmission of HCMV, that is via the placenta from infected mother to fetus, is one of the leading causes of congenital sensorineural hearing loss (SNHL) and neurodevelopmental complications in newborns. It accounts for up to 10% of SNHL cases in newborns, of both unilateral and bilateral types. Antiviral therapy is helpful in such cases if administered within the first month of life, and for hearing impairment, cochlear implants have been used to treat children who develop profound hearing loss. The child can present with hearing loss at birth or it may be late in onset and of progressive type. Suspected children should be evaluated regularly for the early detection of hearing loss and to provide the appropriate treatment.

Congenital Cytomegalovirus and Hearing Loss: The State of the Art.

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In developed countries, congenital cytomegalovirus (cCMV) infection is the most common congenital viral infection, representing the leading non-genetic cause of sensorineural hearing loss (HL). Diagnosis of cCMV infection can be performed by detection of CMV DNA in urine or saliva within 2-3 weeks after birth, or later in dried blood samples on the Guthrie card. Currently, there are many controversies regarding the preventive, diagnostic, and therapeutic approaches to cCMV infection. HL secondary to cCMV is highly variable in onset, side, degree, audiometric configuration, and threshold changes over time. Therefore, it is of paramount importance to perform a long and thorough audiological follow-up in children with cCMV infection to ensure early identification and prompt treatment of progressive and/or late-onset HL. Early cochlear implantation appears to be a valid solution not only for children with bilateral profound HL, but also for those with single-sided deafness, improving localization ability and understanding speech in noisy environments. Moreover, the decision to apply a unilateral cochlear implant in children with cCMV is strengthened by the non-negligible possibility of hearing deterioration of the contralateral ear over time.

Quantification of human cytomegalovirus by competitive PCR and capillary electrophoresis.

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Human cytomegalovirus (CMV) is a ubiquitous pathogen found in 40-100% of adults, and in about 1% of live births in the United States (1). It is the most common fetal and perinatal infectious organism; approx 10% of infected neonates are born with symptomatic congenital CMV disease, which is the most common cause of mental retardation and childhood deafness. CMV is a significant pathogen in immunocompromised individuals, including organ transplant recipients (2-4), and in acquired immune deficiency syndrome (AIDS) patients (5,6). Infection is characterized by latency, chronic infection, and reactivation, a progression similar to that observed in other members of the herpesvirus family. Because CMV infection is usually controlled by the host cellular immune system, primary infections can occur without obvious symptoms, and progress to latency may go unnoticed. Latent infection may persist throughout life, but primary or reactivated infection, coupled with impaired host immune response, can rapidly produce symptomatic CMV disease.

A Novel Non-Replication-Competent Cytomegalovirus Capsid Mutant Vaccine Strategy Is Effective in Reducing Congenital Infection.

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Congenital cytomegalovirus (CMV) infection is a leading cause of mental retardation and deafness in newborns. The guinea pig is the only small animal model for congenital CMV infection. A novel CMV vaccine was investigated as an intervention strategy against congenital guinea pig cytomegalovirus (GPCMV) infection. In this disabled infectious single-cycle (DISC) vaccine strategy, a GPCMV mutant virus was used that lacked the ability to express an essential capsid gene (the UL85 homolog GP85) except when grown on a complementing cell line. In vaccinated animals, the GP85 mutant virus (GP85 DISC) induced an antibody response to important glycoprotein complexes considered neutralizing target antigens (gB, gH/gL/gO, and gM/gN). The vaccine also generated a T cell response to the pp65 homolog

(GP83), determined via a newly established guinea pig gamma interferon enzyme-linked immunosorbent spot assay. In a congenital infection protection study, GP85 DISC-vaccinated animals and a nonvaccinated control group were challenged during pregnancy with wild-type GPCMV (10(5) PFU). The pregnant animals carried the pups to term, and viral loads in target organs of pups were analyzed. Based on live pup births in the vaccinated and control groups (94.1% versus 63.6%), the vaccine was successful in reducing mortality ($P = 0.0002$). Additionally, pups from the vaccinated group had reduced CMV transmission, with 23.5% infected target organs versus 75.9% in the control group. Overall, these preliminary studies indicate that a DISC CMV vaccine strategy has the ability to induce an immune response similar to that of natural virus infection but has the increased safety of a non-replication-competent virus, which makes this approach attractive as a CMV vaccine strategy. Congenital CMV infection is a leading cause of mental retardation and deafness in newborns. An effective vaccine against CMV remains an elusive goal despite over 50 years of CMV research. The guinea pig, with a placenta structure similar to that in humans, is the only small animal model for congenital CMV infection and recapitulates disease symptoms (e.g., deafness) in newborn pups. In this report, a novel vaccine strategy against congenital guinea pig cytomegalovirus (GPCMV) infection was developed, characterized, and tested for efficacy. This disabled infectious single-cycle (DISC) vaccine strategy induced a neutralizing antibody or a T cell response to important target antigens. In a congenital infection protection study, animals were protected against CMV in comparison to the nonvaccinated group (52% reduction of transmission). This novel vaccine was more effective than previously tested gB-based vaccines and most other strategies involving live virus vaccines. Overall, the DISC vaccine is a safe and promising approach against congenital CMV infection.

Diagnosis of congenital CMV infection via DBS samples testing and neonatal hearing screening: an observational study in Italy.

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Congenital Cytomegalovirus (cCMV) is the most common cause of non-genetic hearing loss in childhood. A newborn hearing screening program (NHSP) is currently running in Italy, but no universal cCMV nor statewide hearing-targeted CMV screening programs have been implemented yet. This observational monocentric study was aimed at estimating the rate of cCMV infections identified by CMV-DNA analysis on Dried Blood Spots (DBS) samples in deaf children identified via NHSP in Northern Italy in the period spanning from 2014 to 2018. Children with a confirmed diagnosis of deafness and investigated for CMV-DNA by nucleic acid extraction and in-house polymerase-chain reaction (PCR) on stored newborns screening cards (DBS-test) were included in this study. Deafness was defined by a hearing threshold ≥ 20 decibel (dB HL) by Auditory Brainstem Responses (ABR); all investigated DBS samples were collected within 3 days of life. Overall, 82 children were included (median age: 3.4 months; lower-upper quartiles: 2-5.3 months; males: 60.9%). Most of them (70.7%) presented bilateral hearing loss with a symmetrical pattern in 79.3% of the cases. ABR thresholds were ≥ 70 dB HL (severe/profound deafness) in 46.5% of children. Among all tested children, 6.1% resulted positive for cCMV. The rate of severe/profound deafness was statistically higher in children with cCMV infection. The addition of DBS-test to the NHSP allowed the identification, in their first months of life, of a cCMV infection in 6.1% of children who had failed NHS. The introduction of a targeted CMV screening strategy could help clinicians in the differential diagnosis and in the babies' management. DBS samples can be considered a "universal newborns biobank": their storage site and duration should be the subject of political decision-making.

A clinical pathologic study of hearing loss in congenital cytomegalovirus infection.

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Cytomegalovirus (CMV) infection is currently reported as the most common cause of congenital viral induced deafness. However, few systematic studies of the audiovestibular sequelae of this infection are present in the literature. A clinical pathologic study was conducted from 1976 to 1982 to evaluate this. Fifty-two pairs of infant and children's temporal bone studied demonstrated no evidence of CMV endolabyrinthitis even in the single case with evidence of extensive congenital CMV infection. Over 2,000 umbilical cord sera were screened to detect asymptomatic CMV infection with an incidence of 0.38% (and slightly greater than 1% when extrapolated to correct for the sensitivity of the method of detection) in a central Pennsylvania study population. No sensorineural abnormalities were detected in five asymptomatic children and 30 control children. However, three out of six (50%) infants, symptomatic at birth and followed to a mean age of 5.5 years, showed significant and progressive sensorineural loss and vestibular deficits.

Congenital Cytomegalovirus Infection.

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Congenital cytomegalovirus (cCMV) infection is a leading cause of hearing loss and neurological disabilities in children, with the disease burden and disabilities due to cCMV greater than many other well recognized childhood conditions. A minority of infants with cCMV will have symptoms at birth. Infants with symptomatic cCMV are at higher risk for sequelae than those born without symptoms. The majority of infants with cCMV are asymptomatic at birth, but 10%-15% will develop hearing loss. Although clinical symptoms can help predict which infants will have sensorineural hearing loss, among asymptomatic cCMV there are currently no predictors of adverse outcome. The identification of a biomarker to identify those at highest risk of sequelae is highly desirable to target interventions to those who could potentially benefit. Because there is increasing rationale for establishing both targeted and universal screening programs for cCMV in the United States and worldwide, this is an urgent priority.